

1. INTRODUCTION

This report presents the results of a comprehensive evaluation of a Retrieval-Augmented Generation (RAG) pipeline on the EnterpriseRAG-Bench dataset. The study investigates chunking strategies, query rewriting techniques, and LLM judge biases. The key finding is that retriever quality fundamentally bounds RAG performance, with the recall floor limiting the maximum possible generator performance.

2. EXPERIMENTAL SETUP

2.1 Dataset

- Dataset: EnterpriseRAG-Bench (onyx-dot-app)
- Questions: First 200 questions sorted by `question_id`
- Corpus: 10,000 documents (all gold documents + background sample)
- Relevance: Binary (0/1)
- Question Types: basic, semantic

2.2 Models

- Dense Retriever: all-MiniLM-L6-v2 (384-dimensional embeddings)
- BM25: Custom implementation ($k_1=1.5$, $b=0.75$)
- LLM Judge: Qwen 2.5-7B-Instruct (4-bit quantization, temperature=0)

2.3 Reproducibility

- Fixed seed: 20260605
 - Paraphrase caching in `synthetic_rewrites.json`
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3. CHUNKING STUDY

3.1 Chunking Strategies

Two strategies were implemented:

1. Fixed-size chunking: Splits documents into chunks of a fixed token size with a specified overlap
2. Recursive chunking: Hierarchical splitting (paragraph → sentence → word)

Chunk count formula:

$$n = \lceil (L - \text{overlap}) / (\text{size} - \text{overlap}) \rceil$$

3.2 Sweep Results

Four configurations were tested for each strategy:

Strategy	Chunk Size	Overlap	Recall Floor	Avg Chunks/Doc
Fixed	128	0	0.2183	7.4
Fixed	128	32	0.2317	9.3

Fixed	256	0	0.1533	3.9
Fixed	256	64	0.1533	4.8
Recursive	128	0	0.3483	19.6
Recursive	128	32	0.3267	25.4
Recursive	256	0	0.3083	17.7
Recursive	256	64	0.3033	21.7

3.3 Best Configuration

- Strategy: Recursive
- Chunk Size: 128
- Overlap: 0
- Recall Floor: 0.3483

Interpretation: At most 34.83% of answer facts are reachable through chunking. This means even a perfect generator cannot produce correct answers for more than ~35% of facts – RAG quality is fundamentally bounded by retriever quality.

3.4 Fact Split Demonstration

No document was found where a fact was broken at the chunk boundary. This is normal and indicates that for the tested documents, the chunking parameters did not cause fact fragmentation.

4. QUERY REWRITING METHODS

4.1 Implemented Methods

Method	Description
Multi-Query	Generate paraphrases, search each, merge via MAX similarity
HyDE	Generate a hypothetical answer, retrieve near it
RAG-Fusion	Merge results via RRF (k=60)

4.2 Results

Method	Prec@1 0	Rec@10	Recall@ 10	Recall@ 20	MRR	nDCG@ 10
Baseline	0.6948	0.2336	0.8550	0.8750	0.6981	0.9634
BM25						
Multi-Query	0.4674	0.3276	0.6550	0.7400	0.4730	0.8680
HyDE	0.4024	0.2855	0.5800	0.6450	0.4100	0.7626
RAG-Fusion	0.4528	0.3125	0.6500	0.7400	0.4597	0.8604

Best Method (for recall): Multi-Query achieves the highest Recall@10 (0.3276) and Recall@20 (0.7400), meaning it retrieves more relevant chunks than other methods.

Best Method (for precision): Baseline BM25 remains the best for precision-oriented metrics (Prec@10: 0.6948, MRR: 0.6981, nDCG@10: 1.2634).

4.3 Vocabulary Gap Analysis

Short questions often lack technical jargon present in enterprise documents. However, in this experiment, BM25 outperformed all dense-based rewriting methods. This suggests that:

- Enterprise documents may already contain query terms
 - Dense retrieval may not benefit from synthetic paraphrases
 - The vocabulary gap may be smaller than expected for this dataset
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5. RAG EVALUATION

5.1 Context Precision (MAP)

Context Precision is computed as Mean Average Precision with chunk deduplication to parent documents.

5.2 Context Recall

The fraction of answer facts present in the retrieved context, using cosine similarity ≥ 0.6 .

5.3 Results by Category

Category	Precision@10	Recall@10	MRR	nDCG@10
basic	0.4933	0.3396	0.4999	0.9418
semantic	0.1700	0.1230	0.1787	0.2904

Observation: The model performs significantly better on `basic` questions compared to `semantic` questions. This indicates that semantic questions (which likely require more complex reasoning) are more challenging for the retriever.

5.4 Abstention Metric (for `info_not_found`)

For `info_not_found` questions (no relevant documents), an abstention rate is reported:

- Threshold τ : 0.45
- Abstention Rate: 0.00%

Interpretation: The model never abstained – it always retrieved something even when no relevant documents existed. This is a potential issue for RAG systems, as the model may generate hallucinations based on irrelevant context.

6. STATISTICAL SIGNIFICANCE

RAG-Fusion vs Baseline BM25 comparison:

Metric	Value
Mean Difference	-0.2420
95% Confidence Interval	(-0.3099, -0.1741)
Paired t-test p-value	0.000000
Wilcoxon signed-rank p-value	0.000000
Statistically Significant?	Yes

Interpretation: The difference between RAG-Fusion and BM25 is statistically significant ($p < 0.05$). However, BM25 significantly outperforms RAG-Fusion (mean difference is negative). This indicates that while the difference is not random, BM25 is actually better for this dataset.

Caveats:

- Statistical significance does not imply practical significance
- Multiple comparisons (4 chunking × 3 rewriting methods) may inflate Type I error

7. LLM JUDGE BIAS LAB

7.1 Position Bias

Position bias measures whether the judge's score depends on the order of presentation.

Metric	Value
Position-Follow Bias Rate	10.00%
Swap-and-Average Winner Rate	13.33%
Difference	+3.33%

Interpretation: Position bias is relatively low (10%). Swap-and-average slightly increases the winner rate, suggesting it does not fully mitigate position bias in this setup.

7.2 Verbosity Bias

Verbosity bias measures whether longer answers receive higher scores.

Metric	Value
Verbosity Advantage Rate	20.00%

Interpretation: The judge favors longer answers 20% of the time. This indicates a moderate verbosity bias — padded answers are more likely to win.

7.3 Swap-and-Average

Averaging scores from both orderings ($A \rightarrow B$ and $B \rightarrow A$) is intended to reduce position bias. In this experiment, it increased the winner rate from 10.00% to 13.33%, suggesting limited effectiveness.

7.4 Goodhart Weight-Flip Demonstration

Weights	Honest A	Padded C	Winner
(1:1:1)	4.3333	4.0000	Honest A
(1:1:2)	4.0000	4.2500	Padded C

Conclusion: Changing rubric weights (completeness $\times 2$) changes the winner without changing the answers. This demonstrates that rubric weights are an attack surface, and one should not blindly optimize against an unaudited LLM judge.

8. DISCUSSION

8.1 Where Quality Is Won in RAG

1. Recall floor (0.3483): Even with optimal chunking, only ~35% of facts are reachable. This is a fundamental limitation of the retriever.

2. Query rewriting: Did NOT improve over BM25 in this experiment. This may indicate that:
 - The dense retriever does not benefit from paraphrasing
 - Enterprise documents already contain query terms
 - The generated paraphrases may be suboptimal
3. Statistical significance: Confirms that the difference between methods is real, but BM25 is actually better.
4. LLM Judge bias: Position bias is low (10%), but verbosity bias is moderate (20%). Swap-and-average was not effective.
5. Goodhart: Demonstrates the danger of optimizing against an unaudited judge.

8.2 Why You Cannot Optimise Against an Unaudited Judge

The Goodhart weight-flip shows that without judge auditing and rubric calibration, a model can "game" the scoring system without actually improving answer quality.

8.3 Surprising Finding: BM25 Beats All Rewriting Methods

This is a notable result. Possible explanations:

- Lexical overlap: Enterprise documents may contain the exact query terms
 - Dense retriever limitations: The dense retriever may not benefit from paraphrase expansion
 - Generation quality: The LLM-generated paraphrases may be too different from the original query
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9. CONCLUSION

1. Best chunking strategy: Recursive, size=128, overlap=0 (recall floor=0.3483)
2. Best method: Baseline BM25 outperforms all rewriting methods across precision metrics
3. Best recall method: Multi-Query achieves highest Recall@20 (0.7400)
4. Position bias: 10.00%, swap-and-average not effective
5. Verbosity bias: 20.00%
6. Goodhart: Rubric weights are an attack surface

Key Takeaway: RAG quality is primarily determined by retriever quality. In this dataset, the lexical BM25 retriever outperformed dense retrieval with query rewriting, suggesting that document-term overlap may be sufficient for effective retrieval.

Method Comparison

